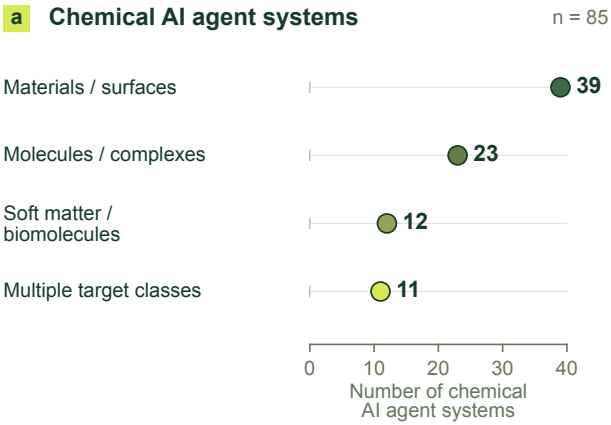
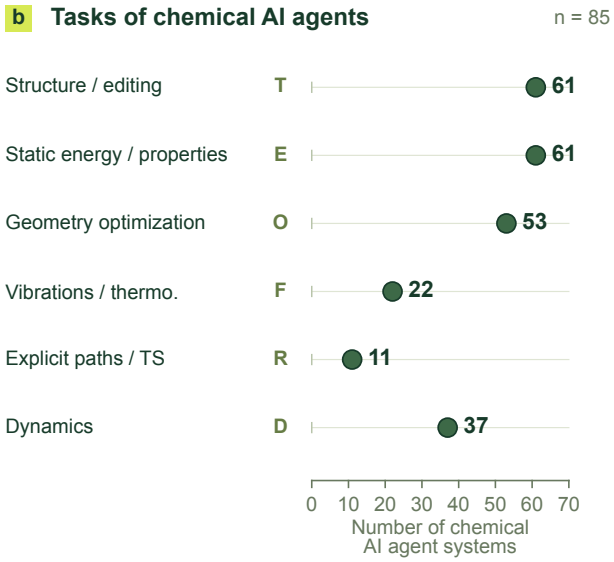

a
Chemical AI agent systems
n = 85
Materials / surfaces
39
Molecules / complexes
23
Soft matter / biomolecules
12
Multiple target classes
11
0
10
20
30
40
Number of chemical AI agent systems


b
Tasks of chemical AI agents
n = 85
Structure / editing
T
61
Static energy / properties
E
61
Geometry optimization
O
53
Vibrations / thermo.
F
22
Explicit paths / TS
R
11
Dynamics
D
37
0
10
20
30
40
50
60
70
Number of chemical AI agent systems

C Materials / surfaces

n = 39 · 1/3

	AI agent system	T	E	O	F	R	D
01	LLaMP	●	○	○	○	○	●
02	ChemReasoner	●	●	●	○	○	○
03	AtomAgents	●	●	●	○	●	○
04	Adsorb-Agent	●	●	●	○	○	○
05	LangSim	●	●	●	○	○	○
06	MDAgent	○	○	○	○	○	●
07	TopoMAS	●	●	○	○	○	○
08	DREAMS	●	●	●	○	○	○
09	SparksMatter	●	●	●	○	○	○
10	VASPilot	○	●	●	○	○	○
11	AMLP	○	●	●	○	○	●
12	GENIUS	○	●	●	○	○	○
13	LAMMPS-Agents	●	●	●	●	○	●
14	AGAPI-Agents	●	●	●	○	○	○

● Demonstrated ○ Unencoded

C Materials / surfaces

n = 39 · 2/3

	AI agent system	T	E	O	F	R	D
15	MASTER	●	●	●	○	○	○
16	VASP Agent	○	●	●	○	●	○
17	Masgent	●	●	●	○	○	○
18	CASCADE	○	●	●	○	○	○
19	MDAgent2-RUNTIME	○	○	○	○	○	●
20	CatMaster	●	●	●	●	●	○
21	El Agente Sólido	●	●	●	●	○	○
22	Catalyst-Agent	●	●	●	○	○	○
23	TritonDFT	○	●	●	●	○	○
24	MatSciAgent	●	○	○	○	○	●
25	SimMOF	●	●	●	○	○	●
26	MatClaw	○	●	○	○	○	●
27	XDFT	●	●	○	○	○	○
28	TSAgent	○	●	●	●	●	○

● Demonstrated ○ Unencoded

C Materials / surfaces

n = 39 · 3/3

	AI agent system	T	E	O	F	R	D
29	Lang2MLIP	●	●	○	○	○	●
30	ChatMOSP	●	○	○	○	○	○
31	CatDT	●	●	●	●	●	○
32	URSA LAMMPS agent	●	●	○	○	○	●
33	AdsMind	●	●	●	○	○	○
34	Agent-MD	○	○	○	○	○	●
35	matmcp	●	●	●	○	○	○
36	PVmatAgent	●	●	●	○	○	○
37	Agent-E2MD	○	●	●	○	○	●
38	MLP Copilot	○	●	○	○	○	●
39	GBClaw	●	●	●	○	●	○

● Demonstrated ○ Unencoded

d Molecules / complexes

n = 23 · 1/2

	AI agent system	T	E	O	F	R	D
01	Buehler molecular DFT	●	●	○	○	○	○
02	El Agente Q	●	●	●	●	○	○
03	Aitomia	●	●	●	●	○	○
04	MatAgent (Lv et al.)	●	●	○	○	○	○
05	ChemGraph	●	●	●	●	○	○
06	AutoSolvate-Agent	●	●	●	●	○	○
07	ChemAgents (azobenzene)	○	○	○	●	○	●
08	El Agente Cuántico	○	●	○	○	○	○
09	ChatMat	●	●	○	○	○	○
10	El Agente Estructural	●	○	●	○	○	○
11	El Agente Quntur	●	●	●	●	●	○
12	Chelatron	●	●	○	○	○	○
13	OpenClaw comp. chemistry	●	○	●	○	○	●
14	Protomia	●	●	○	●	○	○

● Demonstrated ○ Unencoded

d Molecules / complexes

n = 23 · 2/2

	AI agent system	T	E	O	F	R	D
15	El Agente Forjador	●	●	●	●	○	●
16	VirtualLab_CC	●	●	●	●	○	○
17	LADeQ	○	●	○	○	○	○
18	Evolving TS workflows	○	○	●	●	●	○
19	ARCHE	●	●	●	●	●	○
20	Multireference QC agent	○	●	○	○	○	○
21	SQD-Agent	○	●	○	○	○	○
22	QAOS / Quantum Agentic Operating System	●	●	●	○	○	○
23	PULSES	●	●	●	○	○	●

● Demonstrated ○ Unencoded

e Soft matter / biomolecules

n = 12

	AI agent system	T	E	O	F	R	D
01	ProtAgents	●	○	○	●	○	○
02	MDCrow	○	○	○	○	○	●
03	NAMD-Agent	●	○	○	○	○	●
04	ToPolyAgent	●	○	○	○	○	●
05	DynaMate (protein–ligand)	●	○	●	○	○	●
06	Electrolyte Agent / Reasoning-to-Simulation	●	○	○	○	○	●
07	PolyJarvis	●	○	○	○	○	●
08	MDAgent (bio, 2026)	●	○	●	○	○	●
09	DynaMate2	●	○	○	○	○	●
10	MDForge	○	○	○	○	○	●
11	MutexaGPT	●	○	○	○	○	●
12	CGMas	●	○	●	○	○	●

● Demonstrated ○ Unencoded

f Multiple target classes

n = 11

	AI agent system	T	E	O	F	R	D
01	DynaMate (Mendible-Barreto et al.)	●	○	●	○	○	●
02	QUASAR	●	●	●	○	○	●
03	El Agente Gráfico	●	●	●	●	○	○
04	FermiLink	○	●	○	○	○	●
05	CatGo	●	●	●	●	●	○
06	AtomisticSkills	●	●	●	○	○	●
07	AutoDFT	○	●	●	●	○	●
08	MLIPilot	○	●	●	○	○	●
09	Paimon	●	○	●	○	○	●
10	El Agente Potente	●	●	●	●	●	●
11	Magnolia	●	●	●	○	○	●

● Demonstrated ○ Unencoded